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ztamrie
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Zelalem Tamrie

Sr Backend Developer and Data Analytics Engineer

About Me

Senior Backend Developer and Data/AI Engineer with 6+ years of experience designing, developing, and optimizing scalable backend systems, financial platforms, data pipelines, and AI-powered applications. Highly skilled in Go, Python, SQL, distributed systems, ETL automation, BI dashboards, and machine learning workflows.
Proven track record building real-time fintech systems, constructing data warehouses, developing RAG-based AI tools, and contributing to academic research in computer vision and NLP. Strong communicator with both engineering and educational expertise, capable of leading complex technical initiatives end-to-end.

LANGUAGE

English
Amharic

EXPERIENCE

Data Analyst Engineer — Oct '24 - Present

HABTech Solutions

Tech: Python, SQL, Power BI, Superset, Airflow, DBT, ETL Pipelines

- Developed **end-to-end ETL pipelines** to ingest, clean, normalize, and transform healthcare datasets, including **DHIS2**.
- Built interactive **Power BI and Superset dashboards,** improving visibility on national health KPIs and facility-level performance.
- Designed analytical models to compute **health indicator formulas**, tracking compliance, reporting delays, and performance gaps.
- Integrated dashboards with internal systems via secure REST APIs.
- Performed data validation, anomaly detection, and standardized reporting for stakeholders.
- Collaborated with Ministry of Health stakeholders to refine KPIs and ensure accurate data flow.

Senior Backend Developer — Jul '23 - Jul '24

Addispay — Addis Pay Financial Technology

Tech: Go, Node.js, Apache Pulsar, PostgreSQL, WebSockets, Docker

- Engineered scalable **financial microservices** for payments, settlements, refunds, and reconciliation.
- Designed and implemented secure, PCI-aligned **checkout pages** used by merchants and partners.
- Developed real-time communication platforms using **WebSockets** for live transaction updates.
- Built high-throughput **messaging pipelines using Apache Pulsar** for distributed event processing.
- Improved backend system reliability and reduced API latency through query optimization and caching.
- Integrated multiple third-party payment processors using secure API gateways.
- Enhanced system observability by adding structured logging, tracing, and error-handling workflows.

Senior Backend Developer — Jul '24 - Oct '24

BM Technology — Addis Ababa

Backend Development: Golang and Postgres

Tech: Go, PostgreSQL, Accounting Systems

- Implemented IFRS/IAS-compliant **financial accounting system**, including:
  - Journal creation & posting
  - Ledger entries & account balances
  - Trial balance & financial statements generation
- Designed and optimized accounting workflows, ensuring auditability and accuracy.
- Built backend modules enabling automatic data validation and financial reporting.

Computer Science Lecturer — Oct '17 - Present

Wollo University — Wollo University

- Deliver undergraduate courses in: ML, NLP, Algorithms, Distributed Systems, Data Structures.
- Supervise senior projects and academic research in AI and software engineering.
- Provide academic advising and mentorship to students.
- Conduct research in machine learning, computer vision, and Ethiopian-language technologies.

Assistant Lecturer — Oct '16 - Nov '17

Oda Bultum University — Oda Bultum University

- Delivered lectures to students.
- Advised and mentored students.
- Participated in community service projects.
- Conducted academic research.

EDUCATION

Master of Science in Computer Science — Jan '18 - Dec '20

Bahir Dar University — Bahir Dar University, Ethiopia

Final Grade: 3.75

Thesis: Amharic Visual Speech Recognition Using Machine Learning and Computer Vision

Bachelor of Science in Computer Science — Jan '12 - Dec '16

Bahir Dar University — Bahir Dar University, Ethiopia

Final Grade: 3.71 | ECTS Credits: 249

- Industrial Project: Online Bus Ticket Reservation Using JSP and Servlet

Certificate — Jun '25 - Sep '25

Kifiya AI Mastery Training Program (SAFEF KAIM) — Machine Learning Engineering, Data Engineering, Financial Analysis

Mastercard Foundation & Kifiya Financial Technology — September 2025

SKILLS

Backend Engineering

Go (primary), Python, Node.js, Laravel, Gin, Fiber

REST APIs, gRPC, WebSockets, Microservices, Clean Architecture

Apache Pulsar (messaging), Protobuf, Distributed Systems

AI / Machine Learning

Machine Learning, Deep Learning, Model Training

Computer Vision (CNNs, image grading, visual speech recognition)

RAG Systems, LangChain, Prompt Engineering

Embeddings (SentenceTransformers), Vector Databases (Chroma)

Data Engineering

ETL/ELT Pipelines, Data Warehousing

Airflow, DBT, Airbyte

Apache Superset, Power BI

Data Modeling, KPI Design, Performance Metrics

Databases

PostgreSQL, MySQL, SQL Server

MongoDB

Snowflake (foundational)

DevOps & Tools

Docker, GitLab CI/CD, Linux, Git

Monitoring & performance optimization

Cloud-ready containerization

Publications

Tamrie, Z. (2022). Amharic Language Visual Speech Recognition Using Hybrid Features. *Published at Abyssinia Journal of Science and Technology*. Retrieved from https://www.ajol.info/index.php/abjst/article/view/226940

Mekuriaw, K., Demil, G., & Tamrie, Z. (2023). Classification and Grading of Cataracts Using a Deep Convolutional Neural Network. *Abyssinia Journal of Engineering and Computing*. Retrieved from https://abjol.org.et/index.php/ajec/article/view/810

PROJECTS

Medical Chatbot

Start Date - End Date

Tech: Flask, LangChain, Gemini LLM, ChromaDB, Docker

GitHub: https://github.com/zelalem-t8/medical-chatbot

- Built RAG pipeline enabling natural-language question answering over medical PDFs.
- Included automatic document ingestion, chunking, vector storage, and semantic retrieval.
- Integrated Gemini LLM for grounded, safe answer generation.
- Exposed REST API for front-end integration.
- Fully containerized for cloud-ready deployment.

AI Recruiter Automation (Technical Recruiter Agent)

Start Date - End Date

Tech: FastAPI, SentenceTransformers, PostgreSQL, Chroma, SendGrid, HuggingFace

GitHub: https://github.com/zelalem-t8/ai-recruiter

- Extract skills from job descriptions and candidate resumes using NLP.
- Generates embeddings and performs **semantic candidate-job matching**.
- Produces ranking with explainability (semantic score + experience + skill overlap).
- Sends tailored outreach emails using **LLM + SendGrid API**.
- Architected a clean and extensible backend for scalable deployment.